

Beneficial intergenerational effects of exercise on brain and cognition: a multilevel meta-analysis of mean and variance

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ABSTRACT

Physical exercise not only helps to improve physical health but can also enhance brain development and cognition. Recent reports on parental (both maternal and paternal) effects raise the possibility that parental exercise may provide benefits to offspring through intergenerational inheritance. However, the general magnitude and consistency of parental exercise effects on offspring is still controversial. Additionally, empirical research has long overlooked an important aspect of exercise: its effects on variability in neurodevelopmental and cognitive traits. Here, we compiled data from 52 studies involving 4786 rodents (412 effect sizes) to quantify the intergenerational transmission of exercise effects on brain and cognition. Using a multilevel meta-analytic approach, we found that, overall, parental exercise showed a tendency for increasing their offspring's brain structure by 12.7% (albeit statistically non-significant) probably *via* significantly facilitating neurogenesis (16.5%). Such changes in neural anatomy go in hand with a significant 20.8% improvement in neurobehaviour (improved learning and memory, and reduced anxiety). Moreover, we found parental exercise significantly reduces inter-individual differences (i.e. reduced variance in the treatment group) in progeny's neurobehaviour by 10.2% (coefficient of variation ratio, lnCVR), suggesting the existence of an individual by intervention interaction. The positive effects of exercise are modulated by several covariates (i.e. moderators), such as the exercised parent's sex, offspring's sex, and age, mode of exercise, and exercise timing. In particular, parental forced exercise is more efficient than voluntary exercise at significantly improving offspring neurobehaviour (26.0%) and reducing its variability (14.2%). We observed larger effects when parental exercise started before pregnancy. However, exercising only during pregnancy also had positive effects. Mechanistically, exercise significantly upregulated brain-derived neurotrophic factor (BDNF) by 28.9%, vascular endothelial growth factor (VEGF) by 35.8%, and significantly decreased hippocampal DNA methylation by 3.5%, suggesting that brain growth factor cascades and epigenetic modifications can moderate the transmission of parental exercise effects. Collectively, by coupling mean with variance effects, our analyses draw a more integrated picture of the benefits that parental exercise has on offspring: not only does it improve offspring brain development and cognitive performance, but it also reduces inter-individual differences in cognition-related traits. We advocate that meta-analysis of variation together with the mean of a trait provides novel insights for old controversies as well as emerging new questions, opening up a new era for generating variance-based hypotheses.

Key words: meta-analysis of variation, hierarchical meta-analysis, heterogeneity of treatment, precision/personalized medicine, hippocampal neuroplasticity, MWM (Morris water maze), physical fitness, treadmill running, forced swimming, wheel running

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I. INTRODUCTION

(1) Benefits of physical exercise

Lifestyle is tightly linked to health. Lifestyle choices can promote poor health but are also effective interventions to mitigate non-communicative diseases, including both metabolic syndromes (e.g. cardiac disease, obesity) (Gutin *et al.*, 2002) and neurological disorders (e.g. cognitive impairment, Alzheimer disease) (Kivipelto, Mangialasche & Ngandu, 2018). In particular, physical exercise is an economically attractive, non-pharmacological, lifestyle intervention for physical wellbeing, as well as for mental health (Hillman, Erickson & Kramer, 2008). Mounting evidence has linked physical exercise to neural and cognitive benefits (Voss *et al.*, 2019). Such benefits have been demonstrated in both humans (Ludyga *et al.*, 2020) and laboratory rodents (Van Praag *et al.*, 2005).

Human studies have mainly concentrated on the positive effects of exercise on cognitive function, measured by neuropsychological tasks; these tasks include the test scores on executive function, memory, and complex attention. For example, school-age children (aged 4–18 years old) with extra physical activity had improved cognitive performance and content memory during language and mathematics testing (Sibley & Etnier, 2003). Using neuroimaging techniques [e.g. structural and functional magnetic resonance imaging (MRI)], recent studies have shown that aerobic exercise resulted in clear alterations of brain structure, for example, increased hippocampus size in humans (Erickson *et al.*, 2011). Animal studies have not only focused on the beneficial effects of exercise on cognitive behaviour, but also revealed potential underlying mechanisms. For example, laboratory rodents exercising with running wheels showed improvements in learning and memory, accompanied by structural, cellular, and physiological changes in the brain (Van Praag *et al.*, 2005).

(2) Intergenerational inheritance

A growing body of evidence suggests that epigenetic changes (changes in gene expression without genetic modification) underpin the beneficial effects of physical activity on the brain and cognition (Fernandes, Arida & Gomez-Pinilla, 2017). A recent study on the intergenerational transmission of the cognitive benefits of exercise moves this point forward (McGreevy *et al.*, 2019), raising the possibility that exercised parents could transmit beneficial effects intergenerationally *via* epigenetic modifications in their F1 progeny's brain (Benito *et al.*, 2018). For example, in rats, exercising parents *via* wheel running resulted in better cognitive performance in their offspring, as demonstrated by improved performance in the Morris water maze task (Akhavan *et al.*, 2008). In addition, rat offspring of exercised parents given an exercise regime also had lower anxiety (Aksu *et al.*, 2012). Further, emerging evidence suggests that parental exercise could boost offspring brain development (e.g. brain size and neurogenesis) (Bick-Sander *et al.*, 2006).

Parental exercise has also been shown to upregulate brain growth factor cascades in the offspring, subsequently affecting brain development and impacting cognitive abilities.

Currently, vascular endothelial growth factor (VEGF) and brain-derived neurotrophic factor (BDNF) are considered to be the principal growth factors that underlie the benefits of exercise on offspring brain development and cognition (Cotman, Berchtold & Christie, 2007). To support brain development, such as neurogenesis, the brain must have sufficient nutrient and energy supply. Such demands are met by increased vascular growth (i.e. growth of new blood vessels) in the hippocampus (Bick-Sander *et al.*, 2006) and cortex (Sanches *et al.*, 2018). VEGF is the neural growth factor promoting the development of new vasculature and facilitating exercise-induced neurogenesis in the central nervous system (Cotman *et al.*, 2007). In parallel, BDNF plays a beneficial role in hippocampal function and cognitive ability (i.e. learning and memory) in exercised rodents. Exercise increases the biosynthesis and expression of BDNF in many brain regions, such as the hippocampus (Cotman & Berchtold, 2002; Horowitz *et al.*, 2020). Several studies have reported that blocking BDNF signalling using antibodies attenuates the positive effects of exercise on the performance of hippocampus-dependent cognitive tasks (Vaynman, Ying & Gomez-Pinilla, 2004; Vaynman *et al.*, 2006).

Despite accumulating evidence for the heritable benefits of exercise, associated findings appear to be inconsistent: several studies of exercised pregnant humans and rodents, for example, failed to detect any benefits on offspring brain development and cognition (Yin *et al.*, 2013; Short *et al.*, 2017; Yeshurun *et al.*, 2017; Benito *et al.*, 2018; Gobinath *et al.*, 2018). There are two major reasons for this inconsistency. First, the sample sizes in these empirical studies were relatively small (offspring $n = 3–40$). Such limited statistical power means that these studies were unlikely to detect intergenerational effects, which are expected to be relatively small (Segabinazi *et al.*, 2019). Second, differences in experimental designs (exercise mode, frequency, duration, intensity), and biological characteristics (e.g. sex of the exercised parent, offspring's sex, and age), could alter the transmission of positive effects. For example, voluntary exercise in parents is predicted to have stronger positive effects on offspring cognition than forced exercise, because the latter is known to induce the release of stress hormones, negatively affecting foetal development (Akhavan *et al.*, 2008). Yet, there is a lack of robust evidence as to whether parental exercise provides widespread benefits for offspring brain development and cognition.

Meta-analytic approaches can overcome power limitations of single studies and quantify general trends, while being capable of exploring any methodological and biological factors driving heterogeneity in effects. Also, meta-analysis allows us to independently explore both the mechanisms and consequences of parental exercise effects on offspring, by allowing one to synthesise concurrently the expression of growth factors in the brain (genes or proteins related to cognition), brain structure (characterized by neural

anatomy) and neurobehaviour performance (measured *via* learning, memory, and anxiety).

(3) Potential moderators

Based on the literature, we identified three types of potential covariates associated with moderating the effects of parental exercise on offspring brain and cognition. First, conventional wisdom is that forced exercise (e.g. treadmill running, forced swimming) acts as an ‘intense stressor’ that can activate the hypothalamic–pituitary–adrenal (HPA) axis and elicit the release of adrenal steroids (Yanagita *et al.*, 2007). Thus, the positive effects of exercise could be reduced or inhibited by stress effects. A multitude of studies have linked prenatal stress to cognitive impairment (Graignic-Philippe *et al.*, 2014). By contrast, animals that are voluntarily exercised (e.g. wheel running) can choose when and how much to exercise (Vaynman *et al.*, 2004), avoiding exercise-associated stress.

Second, the effects of exercise and/or stress on offspring may be dependent on both the sex of the exercised parent as well as the sex of the offspring. Although male and female mammals contribute genetic information equally to progeny (Wells, 2014), females have a stronger non-genetic influence on offspring development. Mammalian mothers can provide not only large amounts of cytoplasmic factors to the egg but also physiological nurturing during pregnancy and lactation (Vik *et al.*, 2014). Fathers contribute little more than sperm during fertilization in mammals (Crean & Bonduriansky, 2014). Therefore, maternal exercise is predicted to have greater benefits on offspring brain development and cognition than paternal exercise (Vik *et al.*, 2014). In addition, there is a sex difference in vulnerability to gestational stress: male foetuses are more highly impacted than female foetuses, termed ‘male vulnerability’ (Mueller & Bale, 2007; Robinson & Bucci, 2014). Therefore, male offspring are expected to experience greater benefits of parental exercise than female offspring.

Third, the timing of exercise and ontogenetic stage of offspring may be important contributing factors. Evidence suggests that very long-term maternal exercise could negatively influence foetal development, rather than acting as a beneficial stimulus which affects physiological (Mastorci *et al.*, 2009), neurodevelopmental (Graignic-Philippe *et al.*, 2014) and behavioural function (Kinney *et al.*, 2008) in the offspring. Parental exercise benefits are expected gradually to diminish as offspring grow from the postnatal stage to adulthood, as cognitive abilities are increasingly shaped by the postnatal environment (Salthouse & Davis, 2006).

(4) Our aims

Given the potential moderators discussed above, our predictions are sixfold: (i) there will be a general and positive trend for intergenerational effects of parental exercise on offspring traits related to brain and cognition; (ii) voluntary exercise by parents is expected to be more beneficial than forced exercise

(Akhavan *et al.*, 2008); (iii) short-duration exercise is likely to result in greater benefits because long-duration exercise is expected to mimic prenatal stress, which negatively impacts foetal development (Graignic-Philippe *et al.*, 2014); (iv) maternal exercise will show stronger benefits than paternal exercise because of the close and extended interaction between mammalian mothers and their developing offspring (Vik *et al.*, 2014); (v) F1 males will experience greater benefits than F1 females, given male vulnerability to gestational stress during development (Mueller & Bale, 2007; Robinson & Bucci, 2014); and (vi) beneficial effects of parental exercise are likely to diminish as offspring age (Salthouse & Davis, 2006). Here, we attempt to test these predictions by implementing a hierarchical meta-analytic approach to a wealth of rodent data collected from the literature.

Importantly, our approach extends beyond quantifying the effects on the mean trait values of the offspring. Statistical analyses have historically focused on whether exercise benefits cognition (effects on the mean; see Fig. 1B–D) (Davenport *et al.*, 2019; Ludyga *et al.*, 2020). However, this overlooks another important question: does parental exercise impact all offspring in a similar way (i.e. does it affect trait variance; see Fig. 1C) (Khouri & Galea, 2016)? We now have meta-analysis of variation to address this second question (Nakagawa *et al.*, 2015). As such, we tested six associated questions: (i) does parental exercise reduce variability in offspring cognitive traits (see Fig. 1D); (ii) does voluntary exercise and forced exercise similarly impact cognitive trait variability in offspring; (iii) are cognitive traits less variable in offspring whose parents are exposed to long- *versus* short-duration exercise; (iv) does variation in cognitive traits depend on whether mothers or fathers are exercised; (v) does an effect on variation in cognitive traits depend on the sex of offspring; and (vi) does parental exercise affect cognitive variability across offspring lifespan?

II. METHODS

(1) General statistical modelling approach

Here, we briefly describe our data collection and statistical modelling approaches for testing the aforementioned predictions (more details on each step are provided in the following sections). First, we searched for relevant studies using online scientific literature databases, including ‘grey’ (non peer-reviewed) literature. We then employed backward- and forward-searching strategies to identify additional literature missed in the main database search. Next, we screened the literature based on pre-defined inclusion criteria. From the included studies, we extracted essential data (e.g. measures of effect size and moderators) to compile the data set for meta-analysis. For statistical modelling, we employed a multilevel linear mixed-effects framework to analyse the collected data. To address our hypotheses and questions, we ran a series of intercept-only models, univariate (uni-moderator)

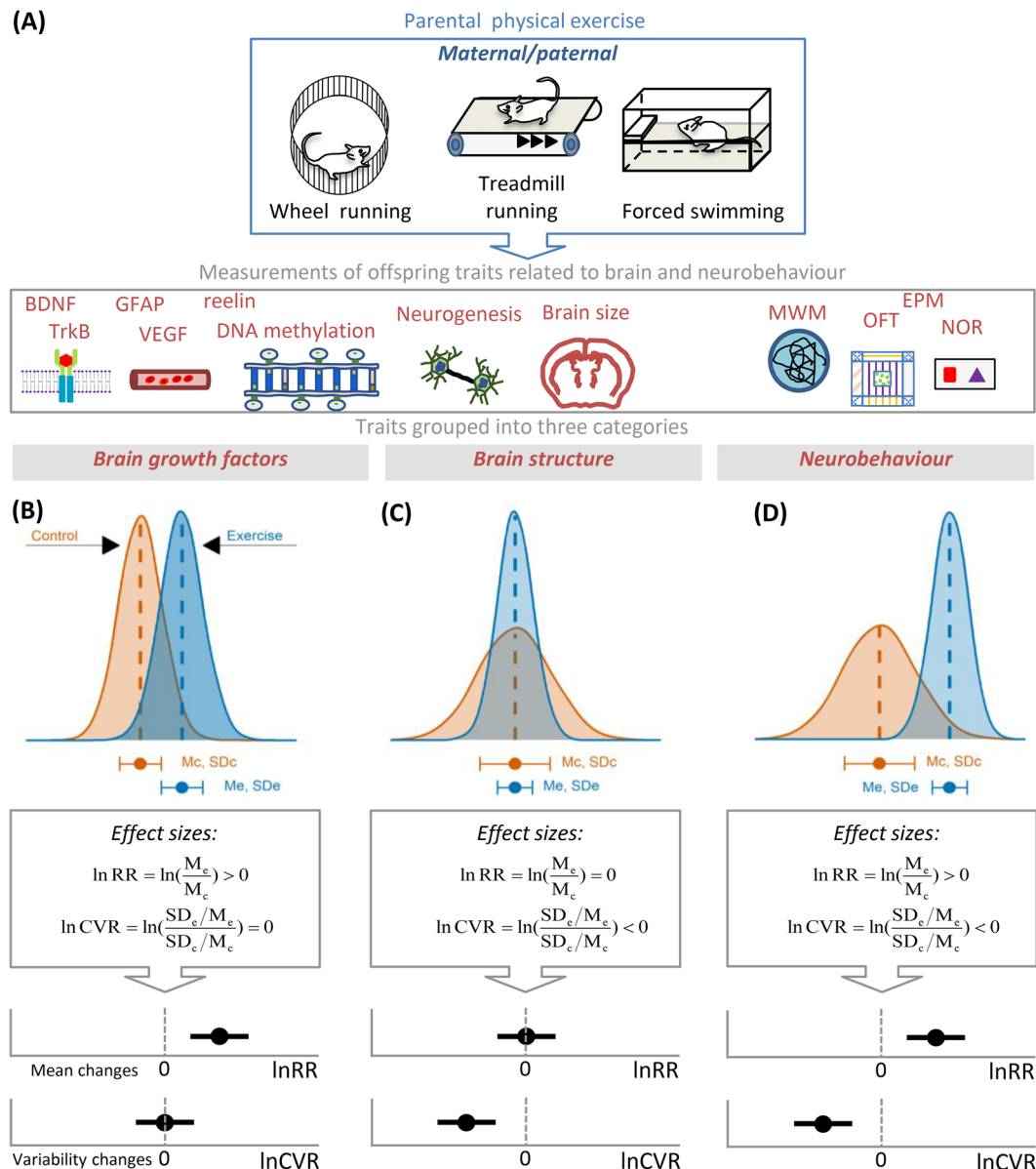


Fig 1. Conceptual diagram displaying three hypothetical outcomes of parental exercise on offspring traits. (A) Three ways of exercising rodent parents: wheel running (including environment enrichment), treadmill running, forced swimming. Three broad categories of offspring traits related to the brain and cognition are shown in the lower panel: growth factors in brain (expression of genes or proteins related to cognition), brain structure (characterized by neural anatomy) and neurobehaviour performance (measured *via* learning, memory, and anxiety). BDNF, brain-derived neurotrophic factor; TrkB, tyrosine kinase receptor B; VEGF, vascular endothelial growth factor; GFAP, glial fibrillary acidic protein; MWM, Morris water maze; OFT, open field test; EPM, elevated plus maze; NOR, novel object recognition. (B) Offspring of exercised parents (blue) on average have greater cognitive trait mean ($\ln RR > 0$) and similar trait variability ($\ln CVR = 0$) than offspring from sedentary parents (orange). (C) Offspring from exercised parents on average have similar cognitive trait mean ($\ln RR = 0$) coupled with less variability ($\ln CVR < 0$). (D) Offspring from exercised parents has greater cognitive trait mean ($\ln RR > 0$) coupled with less variability ($\ln CVR < 0$). We employed a multilevel meta-analytic approach to detect differences in cognitive trait mean ($\ln RR$) and variance ($\ln CVR$) – the two types of effect sizes that can be used for comparing exercised and control groups. $\ln RR$ is the natural logarithm of response ratio between means in the exercise and control group. $\ln RR > 0$ implies that parent exercise improves offspring cognitive trait mean (e.g. cognition and brain structure) or upregulates gene/protein expression in the brain (growth factors). $\ln CVR$ is the natural logarithm of the coefficients of variation ratio (i.e. SD divided by mean) for the exercised and control groups. $\ln CVR < 0$ implies that parent exercise reduces the inter-individual differences in traits. M_e and M_c are the mean values of cognition-related traits for offspring of exercised and control parents, respectively; SD_e and SD_c are the standard deviation of cognition-related traits for offspring of exercised and control parents, respectively.

models, and multi-moderator models. Finally, to examine the robustness of these models, we performed model selection, Egger regression, and time-lag analysis. We performed all analyses in the R environment v.3.6.1 (R Development Team, 2019). All data sets and code used in this study are available at https://github.com/Yefeng0920/Intergenerational_effect_of_exercise_MA_Git.

(2) Data collection

(a) Identification

We followed PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines (Moher *et al.*, 2009) for reporting our systematic literature search and data collection. Figure 2 presents a PRISMA diagram, depicting our processes of identifying and screening articles. Briefly, on April 12, 2018, we conducted searches of the *SCOPUS* and *Web of Science* (WoS) databases for published articles (the exact search strings are listed in Appendix S1). We updated our literature search on October 17, 2019, following the same procedures. To identify any literature missed in the main database searches, we carried out backward and forward reference searching (i.e. a 'snowballing' search). To alleviate a potential file-drawer problem, we also actively searched for relevant grey literature (e.g. theses, reports) using the online databases *ProQuest* and *EBSCO*. After removing duplicates, our systematic searches yielded 905 articles for screening (842 from *SCOPUS* and *WoS*, 5 from snowballing, and 58 from grey literature searches; see Fig. 2).

(b) Screening

We conducted screening in two steps: abstract screening and full-text screening. Two reviewers (Y.Y. and Y.Z.F.) independently screened all 905 identified articles using pre-defined inclusion criteria (see decision trees in Figs S1 and S2). After each screening step, they resolved all conflicting decisions by consensus or consultation with two other reviewers (M.L. and S.N.). In short, the included studies had to: (i) be empirical studies on wild-type laboratory rodents; (ii) expose female or male rodents to exercise intervention; and (iii) report measurements of traits related to brain and cognition of the offspring in a form usable for calculation of effect sizes (see Figs S1 and S2). A list of excluded studies (with reasons for exclusion) following full-text screening is presented in Table S1, and a list of the studies included is provided in Table S2.

(3) Data set compilation

(a) Data extraction

For each included study, we extracted four types of data, including: (i) descriptive statistics (e.g. mean, standard deviation, sample size) for each offspring trait of interest; (ii) animal cohort characteristics (e.g. exercised parent's sex, offspring sex and age); (iii) qualitative (exercise mode) and quantitative

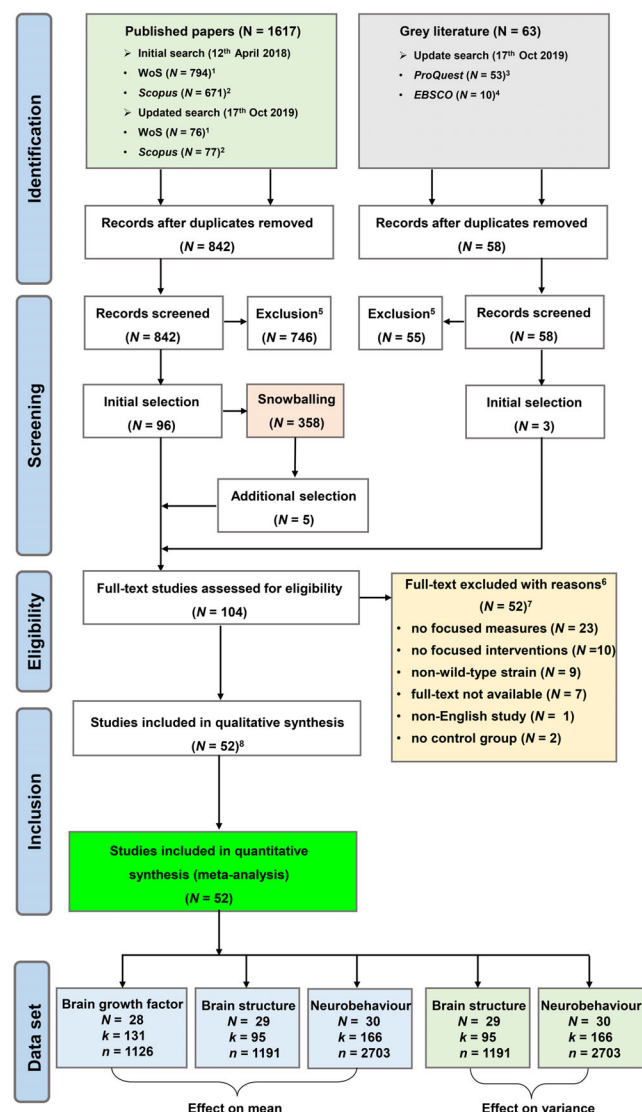


Fig 2. PRISMA flowchart detailing the data-collection process. *N*, number of studies remaining after each stage of the selection process; *k*, number of effect sizes; *n*, number of animals in studies. ^{1–4}The exact search strings and search algorithms for *SCOPUS*, *Web of Science* (WoS), *ProQuest*, and *EBSCO* are listed in the Supporting Information, Search strings section; ⁵title and abstract screening decision-tree is presented in Fig. S1; ⁶full-text screening decision-tree is presented in Fig. S2; ⁷excluded studies, with reasons for exclusion, are listed in Table S1; ⁸included studies are listed in Table S2.

exercise characteristics (frequency, duration, intensity); and (iv) publication characteristics (e.g. publication year, title, journal). Table S3 provides a list and descriptions of all extracted and coded variables (meta-data).

We used the R package *metaDigitise* v.1.0 (Pick, Nakagawa & Noble, 2019) to digitize descriptive statistics from Figs Y.Y. extracted and coded the data, and two other authors (M.L. and Y.Z.F.) carried out cross-checking of the extractions (20% of papers). We validated the accuracy of

the extracted data using a mixed model approach (Stoffel, Nakagawa & Schielzeth, 2017), which indicated that our extractions were completely repeatable (intra-class correlation $ICC = 1$ and inter-class correlation $r = R^2 = 1$; see Figs S3 and S4 and Table S4 for further details).

(b) Coding

In brief, we used three types of coded data items in statistical modelling: (i) identifiers for controlling the sources of variance in our statistical modelling (i.e. trait category, strain of rodent, and between-study and within-study identifiers); (ii) predictors (i.e. moderators), such as the exercised parent's sex, offspring sex and age at measurement, and exercise timing, mode and intensity; and (iii) trait categories (in subset analyses). We categorized the 15 separate offspring cognition-related traits into three broad trait categories: (i) brain growth factors (gene/protein expression), which included BDNF, tyrosine kinase receptor B (TrkB), VEGF, glial fibrillary acidic protein (GFAP), reelin and synapsin levels; (ii) brain development, which included measures of neurogenesis and brain size measurements; (iii) neurobehaviour, which included cognitive behaviour [learning–MWM (Morris water maze), memory–MWM, memory–NOR (novel object recognition)] and anxiety behaviour [anxiety–EPM (elevated plus maze), anxiety–OFT (open field test)]. We have two notes. Firstly, anxiety behaviour is an important predictor of cognitive performance. Therefore, it is sensible to synthesize the two types of behaviours (cognition and anxiety) together to reveal general trends of parental exercise on neurobehaviour. In particular, we employ a multilevel framework, which allows us to combine different types of traits by accounting for different random effect structure (see Section II.4b for detailed explanation). Secondly, although global DNA methylation was measured as a type of gene/protein expression, we treated it separately in the statistical analysis as it does not belong biologically to brain growth factors. Aggregating outcome measurements into these three categories enabled a powerful approach (i.e. sufficient sample size) to reveal how intergenerational effects of parental exercise impact key cellular, structural, and physiological processes that are hypothesized to be important to brain development, learning, and memory. In other words, it allowed us to identify types of changes in the brain that were most widely impacted, providing an indication of mechanisms underlying changes in measured cognitive traits. When we incorporated trait categories into multilevel linear mixed-effects meta-analysis as a random factor, we found that only very small amounts of variance were explained by our trait categories (see Section IV.2d). This approach corroborated the rationality of our trait classifications.

(4) Statistical modelling

(a) Effect size statistics

To test how offspring mean phenotype changes with parental exercise, we calculated the log response ratio (lnRR; Hedges, Gurevitch & Curtis, 1999), which is defined as follows:

$$\ln RR = \ln \left(\frac{M_e}{M_c} \right), \quad (1)$$

$$s_{\ln RR}^2 = \frac{s_e^2}{n_e M_e^2} + \frac{s_c^2}{n_c M_c^2}, \quad (2)$$

where M_e and M_c are the mean measurement values of traits related to brain and cognition for offspring from exercised (e) and control (c) parents, s_e^2 and s_c^2 are the variances of M_e and M_c , and n_e and n_c the sample sizes for offspring from exercised (e) and control (c) parents. The lnRR represents the natural logarithm of the ratio between trait means of offspring of exercised parents and their control counterparts. As higher trait values are indicative of better cognitive performance, positive values of lnRR imply that the offspring from exercised parents benefit more than their control counterparts (Fig. 1D). Note that we flipped the sign of the effect size when lower trait values (e.g. anxiety) indicated better performance so that the sign of effect size from each trait can be interpreted consistently. To aid interpretation, we exponentiated lnRR values as a percentage of the difference between the exercise and control groups.

To test how phenotypic variability changes with parental exercise, we calculated the log coefficient of variation ratio (lnCVR) (Nakagawa *et al.*, 2015), as follows:

$$\ln CVR = \ln \left(\frac{CV_e}{CV_c} \right) + \frac{1}{2(n_e - 1)} - \frac{1}{2(n_c - 1)}, \quad (3)$$

$$s_{\ln CVR}^2 = \frac{s_c^2}{n_c M_c^2} + \frac{1}{2(n_c - 1)} - 2\rho_c \sqrt{\frac{s_c^2}{n_c M_c^2} \frac{1}{2(n_c - 1)}} + \frac{s_e^2}{n_e M_e^2} + \frac{1}{2(n_e - 1)} - 2\rho_e \sqrt{\frac{s_e^2}{n_e M_e^2} \frac{1}{2(n_e - 1)}}, \quad (4)$$

where CV_e and CV_c are the coefficient of variation for offspring from exercised (e) and control (c) parents, and ρ_e and ρ_c are correlations between the logged means and standard deviation of exercised and control groups (note, these correlations are assumed to be 0; see Senior, Viechtbauer & Nakagawa, 2020). All other notations are as described above.

lnCVR represents the ratio between the coefficients of variation (i.e. standard deviation divided by mean) for the offspring of exercised parents and their control counterparts. In this instance, negative lnCVR values would imply that for a given trait related to brain and neurobehaviour, parental exercise reduces inter-individual differences in offspring traits (Fig. 1D). In other words, positive values of lnCVR imply increased variability in the traits of offspring of exercised animals in comparison to their control counterparts (i.e. more variable performance).

The calculation of lnCVR for brain growth factor traits (genes and proteins) involves the variance (or standard deviation) for the expression of reference genes. However, we note that the included studies rarely reported variance for the expression of reference genes, which prevented us from

calculating lnCVR for brain growth factor. Therefore, we do not report the results of lnCVR for this trait in the main text. Instead, we estimated lnCVR for brain growth factor with a strong assumption: the variance in reference gene expression is negligible. This allowed us to calculate lnCVR for brain growth factor although this assumption is unlikely to be true. Nonetheless, we report the results of lnCVR for brain growth factors in Table S11.

(b) Multilevel linear mixed-effects framework

We adopted a multilevel linear mixed-effects framework to fit our statistical models (Nakagawa & Santos, 2012). Multiple effect sizes came from the same study, and many were derived from the same animals within a study, leading to a number of sources of non-independence that we dealt with in our models (Noble *et al.*, 2017). Multiple traits were measured on the same animals within a study which induces a correlation between sampling error variances (Noble *et al.*, 2017). Thus, we constructed variance–covariance matrices to model shared sampling error for effect sizes from the same cohort, assuming a medium correlation (0.5) between sampling variances within the same cohort (Noble *et al.*, 2017).

To account for the hierarchical structure of our data, and deal with additional sources of non-independence, we fitted models with a series of random effects. We employed an information-theoretic approach to decide the most appropriate random effect structure (i.e. multilevel models) (Gurka, 2006). We compared null models (intercept-only), which were run using the maximum likelihood (ML) estimation (rather than restricted maximum likelihood, REML), with permutations of three random factors: (i) trait category identity; (ii) rodent strain; and (iii) study. We also included a residual (observation-level) random effect in our models. The information-theoretic approach used allowed us to examine the fit and plausibility of various models, focusing on models that contain none, one, and up to four (i.e. all) of these random factors. By comparing the variance components and Akaike Information Criteria (AIC), we decided on the appropriate random-factors structure, as reported in Tables S5–S10. We chose a multilevel linear mixed-effects model with the identities of strain, study, and effect size (equivalent to residual-level in a normal linear mixed model), as the random factors.

(c) Intercept-only models

To estimate the overall (average) effect of parental exercise on offspring brain development and cognition, we first fitted meta-analytic (intercept) models (for each of the data subsets representing brain and cognition-related offspring trait categories, for lnRR and lnCVR, independently), as follows:

$$y_{j[i]} = \beta_0 + s_{k[i]} + u_{j[i]} + e_{j[i]} + m_{j[i]}, \quad (5)$$

$$\mathbf{s} \sim \text{Gaussian}(\mathbf{0}, \sigma_s^2 \mathbf{I}) \quad (6)$$

$$u \sim \text{Gaussian}(0, \sigma_u^2 \mathbf{I}), \quad (7)$$

$$e \sim \text{Gaussian}(0, \sigma_e^2 \mathbf{I}), \quad (8)$$

$$m \sim \text{Gaussian}(0, M), \quad (9)$$

Where $y_{j[i]}$ denotes the i th effect size in the j th study, β_0 denotes the overall mean (i.e. intercept for lnRR or lnCVR), $s_{k[i]}$ is the strain-specific effect for the k th strain applied to the i th effect size estimate, $\mathbf{s}$ is a $N_{\text{study}} \times 1$ vector of $s_{k[i]}$, which is normally distributed around 0 with the strain-specific variance σ_s^2 , $\mathbf{I}$ is an identity matrix, $u_{j[i]}$ is the study-specific effect of the j th study applied to the i th effect size, $\mathbf{u}$ is a $1 \times N_{\text{study}}$ vector of $u_{j[i]}$, which is normally distributed around 0 with the between-study variance σ_u^2 , $e_{j[i]}$ is the within-study (effect size) effect for the i th effect size, $\mathbf{e}$ is a $N_{\text{study}} \times 1$ vector of $e_{j[i]}$, which is normally distributed around 0 with the within-study variance σ_e^2 , $m_{j[i]}$ is a sampling error effect for i th effect size, $\mathbf{m}$ is a $N_{\text{study}} \times 1$ vector of $m_{j[i]}$, which is normally distributed around 0 with the corresponding sampling error variance σ_i^2 , $\mathbf{M}$ is a $N_{\text{study}} \times N_{\text{study}}$ matrix with its diagonal elements being σ_i^2 .

To address our main questions, we ran intercept-only models on three general trait categories (i.e. brain growth factors, brain structure, and neurobehaviour – learning, memory, and anxiety), and their specific traits, for lnRR and lnCVR effect sizes, independently, using Equations 5–9. When estimating the effects on brain growth factors, we did not include global DNA methylation in the brain growth factors category, because global DNA methylation is not biologically a brain growth factor. Instead, we fitted models for DNA methylation separately.

(d) Univariate regression models

Our meta-regression models were:

$$y_{j[i]} = \beta_0 + \sum_{h=1}^p \beta_h x_{hj[i]} + s_{k[i]} + u_{j[i]} + e_{j[i]} + m_{j[i]}, \quad (10)$$

where $x_{j[i]}$ is the i th value of the j th study for the h th predictors, the i th predictor (i.e. moderator) from the j th study, β_h is the regression coefficient (i.e. slope for lnRR or lnCVR) for h th predictor, and the other notations are as described above.

Apart from our main questions (Question i, see Section I.4), we tested additional questions (Questions ii–vi), by running univariate regression models with pre-planned predictors (i.e. moderators, see below), using Equations 5–10. We added each of the tested moderator variables as a fixed factor to assess their effects in moderating the benefits of parental exercise on traits related to brain and cognition. Our moderators included: (i) sex of parents subjected to exercise (maternal or paternal exercise); (ii) the sex of measured offspring (male, female, mixed-sex); (iii) the mode of parental

exercise (forced, voluntary); (iv) the timing of parental exercise (pre-conception, pregnancy, pre-conception to pregnancy, pre-conception to *post-partum*); and (v) ontogenetic stage of offspring at trait measurement (prenatal, early-postnatal, adolescence, adulthood). For regression models, we quantified the amount of variance explained by each moderator (in response to parental exercise) by calculating its marginal R^2 (*sensu* Nakagawa & Schielzeth, 2013). To obtain the estimates for each level of pre-planned predictor, we fitted (non-contrast) meta-regression models to each of the data subsets representing the given trait category (i.e. brain growth factors, brain structure, neurobehaviour), for lnRR and lnCVR effect sizes, independently. To compare the differences between the estimates for different levels of pre-planned predictor, we fitted contrast meta-regression models to each of the data sets.

(e) Publication bias

Importantly, the validity of estimates from our analyses relies on the assumption of a negligible file drawer problem (publication bias; Rosenthal, 1979), where small and non-significant effects are less likely to be published than those showing larger and significant effects. Publication bias can potentially inflate overall meta-analytic estimates. A funnel plot, with effect sizes plotted against their precision (the inverse of sampling standard error), is commonly used to detect publication bias by visually elucidating funnel asymmetry. However, when significant heterogeneity exists in the data set, heterogeneity can mask or create funnel asymmetry (Nakagawa & Santos, 2012). To tackle this issue, we assessed potential publication bias using two main approaches. First, we examined funnel asymmetry of the 'meta-analytic' residuals (*sensu* Nakagawa & Santos, 2012) plotted against 'precision' (i.e. the inverse of standard error). Second, we adopted Egger's regression to test deviations from funnel asymmetry (Nakagawa & Santos, 2012). Also, we carried out a regression-based test for time-lag bias (a type of publication bias; Jennions & Møller, 2002), to check whether there was a temporal decline in average effect size (by adding publication year as a moderator variable in Equations 6–10). We conducted parallel assessments of publication bias for data subsets for the three general categories of traits (i.e. brain growth factors, brain structure, cognition).

(f) Approach for between-study correlations

Empirical studies suggest that the expression of brain growth factors (e.g. BDNF) correlates both with improved cognitive behaviour and brain development (e.g. neurogenesis) (Cotman *et al.*, 2007). Therefore, we also investigated correlations between expression of brain growth factors, neurobehaviour performance, and brain development. In the context of meta-analysis, we use study-level correlations as we could not calculate individual-level correlations, but we assumed that these two types of correlations would show similar

patterns. Because our data set contained multiple measurements (see Section II.4b), using all effect sizes will miscalculate the study-level correlation (Dingemans, Dochtermann & Nakagawa, 2012). Instead, we used representative effect sizes from each study to calculate correlations. We used three approaches to estimate between-study correlations: maximum, averaged, and minimum effect sizes.

(g) Interpretation of results

We present point estimates (intercept $\beta_{intercept}$ or slope β_{slope}) and their 95% confidence intervals (CIs) from all our statistical models. We refer to estimates as statistically significant if associated CIs did not span across zero (which corresponds to a P -value less than 0.05; Nakagawa & Cuthill, 2007).

III. RESULTS

(1) Description of data set

We included 52 studies, reporting data from 4786 rodents. We extracted 412 effect sizes comparing the difference in traits related to brain development and cognition between 99 groups of offspring from parents exposed to exercise and a control sedentary lifestyle treatment. Of the included effect sizes, 82.3% were from rats (number of effect sizes $k = 339$) and 17.7% from mice ($k = 73$). Rat studies included three major strains, including 28 Long-Evans, 23 Sprague–Dawley and 288 Wistar. Mouse studies included 73 C57BL/6 (i.e. C57BL/6J and C57BL/6N). The included studies differed from each other in experimental design (exercise mode and duration) and biological characteristics (e.g. sex of exercised parent, sex of tested offspring, and ontogenetic stage). These differences were used as potential moderators to explain the heterogeneity of the observed results across studies. There were two types of exercise (forced or voluntary exercise), of which forced exercise studies contributed 60.7% ($k = 250$) effect sizes and voluntary exercise studies contributed 39.3% ($k = 162$) effect sizes to our data set. Exercise duration ranged from 7 to 156 days (mean = 33.9, median = 22.0, SD = 22). Biological characteristics were distributed as follows: (i) two types of sex of exercised parent: 82.5% female ($k = 340$), 17.5% male ($k = 72$); (ii) three categories of tested offspring sex: 15.5% female ($k = 64$), 53.6% male ($k = 221$), 30.8% mixed sex ($k = 127$); (iii) four categories of parental exercise timing, with 26.9% occurring pre-conception ($k = 111$), 49.8% occurring during pregnancy ($k = 205$), 18.9% taking place from pre-conception to pregnancy ($k = 78$) and 4.4% from pre-conception to *post-partum* ($k = 18$); (iv) four types of ontogenetic stages of offspring at trait measurement: 6.1% prenatal stage ($k = 25$), 15.5% early postnatal stage ($k = 64$), 54.4% adolescence ($k = 224$), 24.0% adulthood ($k = 99$).

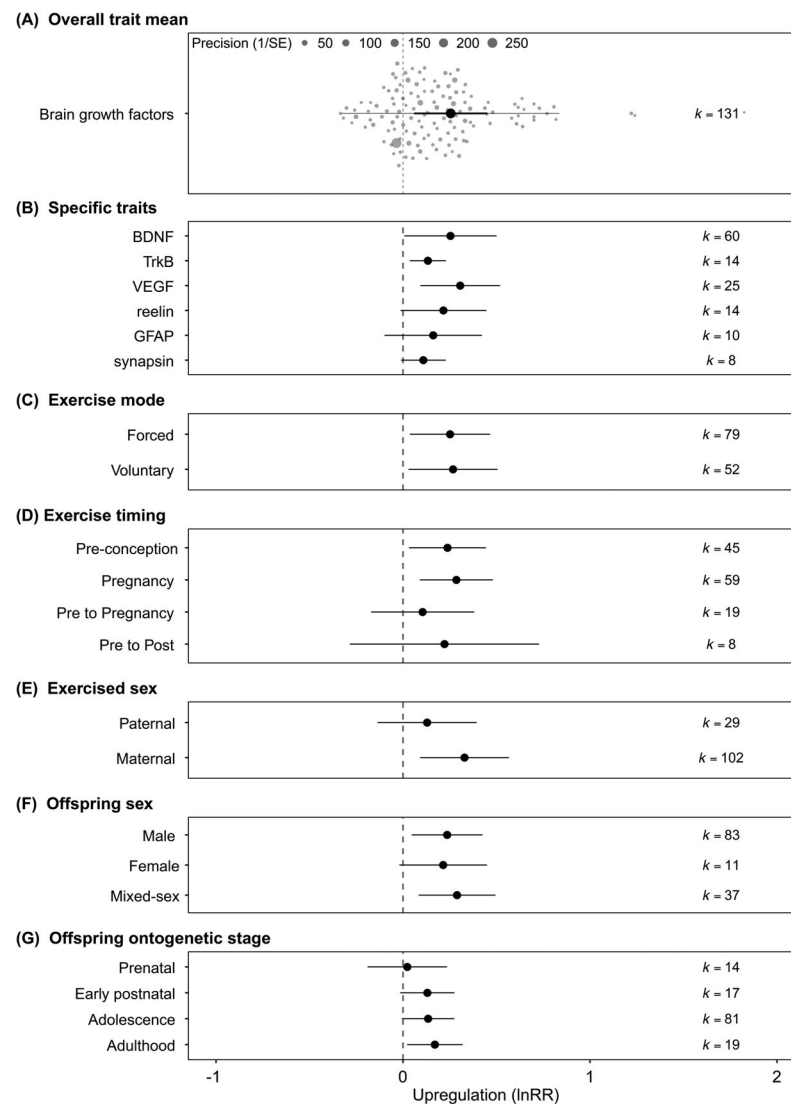


Fig 3. Meta-analytic and meta-regression results for expression of brain growth factors and the more specific trait categories (BDNF, TrkB, VEGF, reelin, GFAP, and synapsin). (A) Orchard (forest-like) plot displaying the overall effect of parental exercise on offspring brain growth factor expression. Black circle and grey circles represent the overall mean differences (lnRR) between exercise and control groups and their individual effect sizes. Thin whiskers denote prediction intervals. Dots signify individual effect sizes scaled proportionally to their precision (precision is the inverse of standard error, SE). k denotes number of effect sizes. (B) Forest plot showing trait means within different trait categories. BDNF, brain-derived neurotrophic factor; TrkB, tyrosine kinase receptor B; VEGF, vascular endothelial growth factor; GFAP, glial fibrillary acidic protein. (C–G) Forest plots displaying offspring brain growth factor expression effect size estimates (lnRR) from univariate regression models with potentially influential moderators: (C) parental exercise mode; (D) parental exercise timing (Pre to Pregnancy, pre-conception to pregnancy; Pre to Post, pre-conception to *post-partum*); (E) exercised sex (parental sex); (F) offspring sex; (G) offspring ontogenetic stage. Circles represent estimates of mean differences between exercise and control groups, and whiskers denote their 95% confidence intervals (CIs). Estimates are statistically significant when their confidence intervals do not span zero (dashed vertical line). lnRR (natural logarithm response ratio) represents the natural logarithm of response ratio between mean responses of gene/protein expression in the exercise and control group. Positive values of lnRR imply that parental exercise improves/upregulates gene/protein expression (e.g. growth factors) in the offspring brain.

(2) Evidence for the intergenerational inheritance of parental exercise effects

(a) Overall trait mean value

Parental exercise triggered the upregulation of offspring brain growth factors by 29.1% ($\beta_{\ln RR} = 0.255$, 95% confidence

interval, CI = 0.059 to 0.451, $k = 131$; Fig. 3A, Table S11), and resulted in a 12.7% improvement (although statistically non-significant) in offspring brain structure ($\beta_{\ln RR} = 0.120$, CI = -0.019 to 0.259, $k = 95$, Fig. 4A, Table S12). We also found that parental exercise improved offspring neurobehaviour by 20.8% ($\beta_{\ln RR} = 0.189$, CI = 0.061 to 0.318,

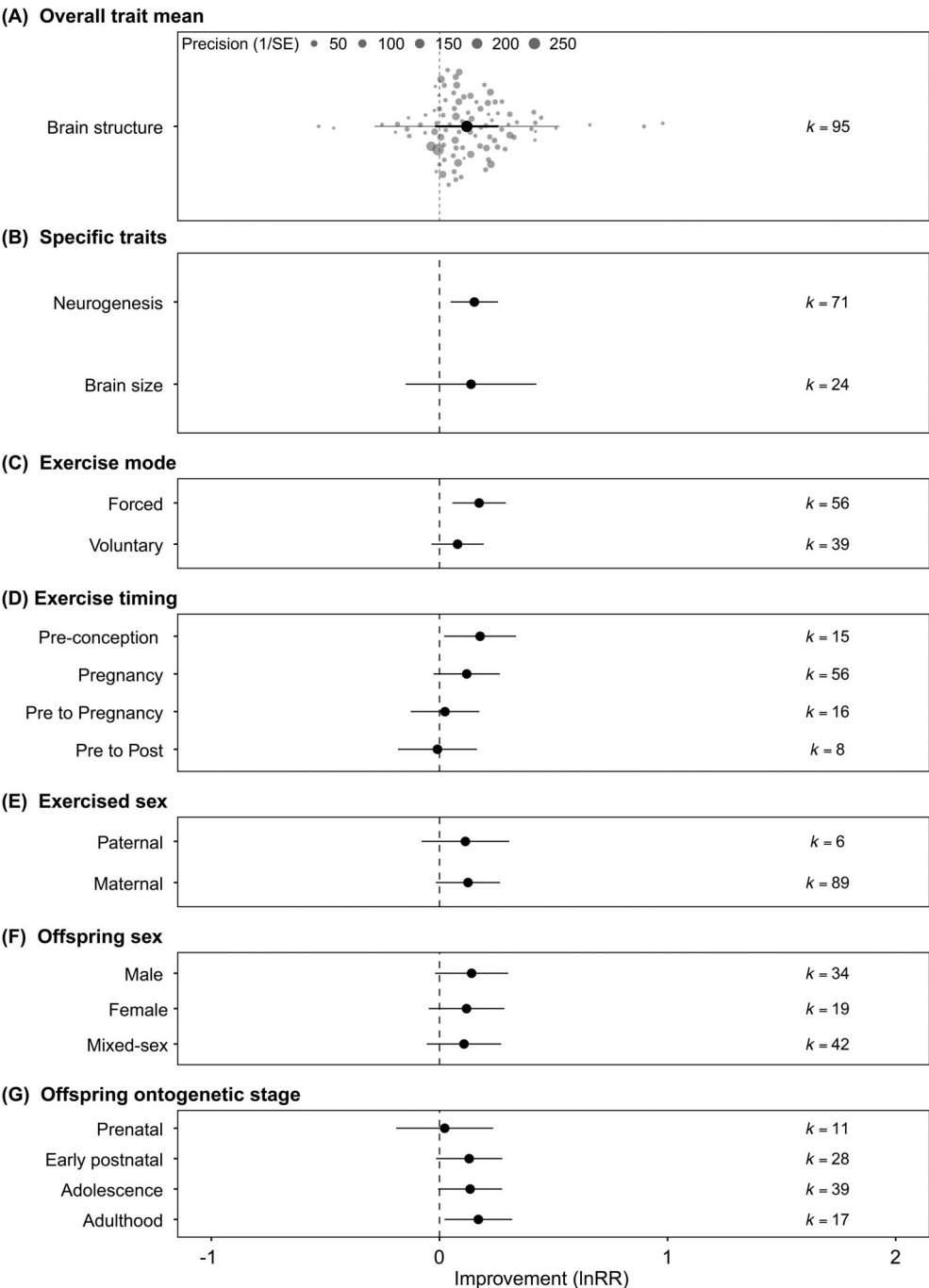


Fig 4. Meta-analytic and meta-regression results for brain structure and the more specific trait categories (brain size and neurogenesis). (A) Orchard (forest-like) plot displaying the overall effect of parental exercise on offspring brain structure. (B) Forest plot showing trait means within different trait categories. (C–G) Forest plots displaying offspring brain structure effect size estimates (lnRR) from univariate regression models with potentially influential moderators: (C) parental exercise mode; (D) parental exercise timing (Pre to Pregnancy, pre-conception to pregnancy; Pre to Post, pre-conception to *post-partum*); (E) exercised sex (parental sex); (F) offspring sex; (G) offspring ontogenetic stage. Positive values of lnRR imply that parental exercise improves offspring brain structure. Other details are as in legend to Fig. 3.

$k = 166$, Fig. 5A and Table S13). Further, parental exercise significantly decreased global DNA methylation in offspring's hippocampus by 3.5% ($\beta_{\ln RR} = -0.035$, CI = -0.049 to -0.022 , $k = 9$, Table S14). There was generally high degree of total heterogeneity ($I^2 = 95.2\text{--}98.3\%$ across all traits, Tables S15–S17).

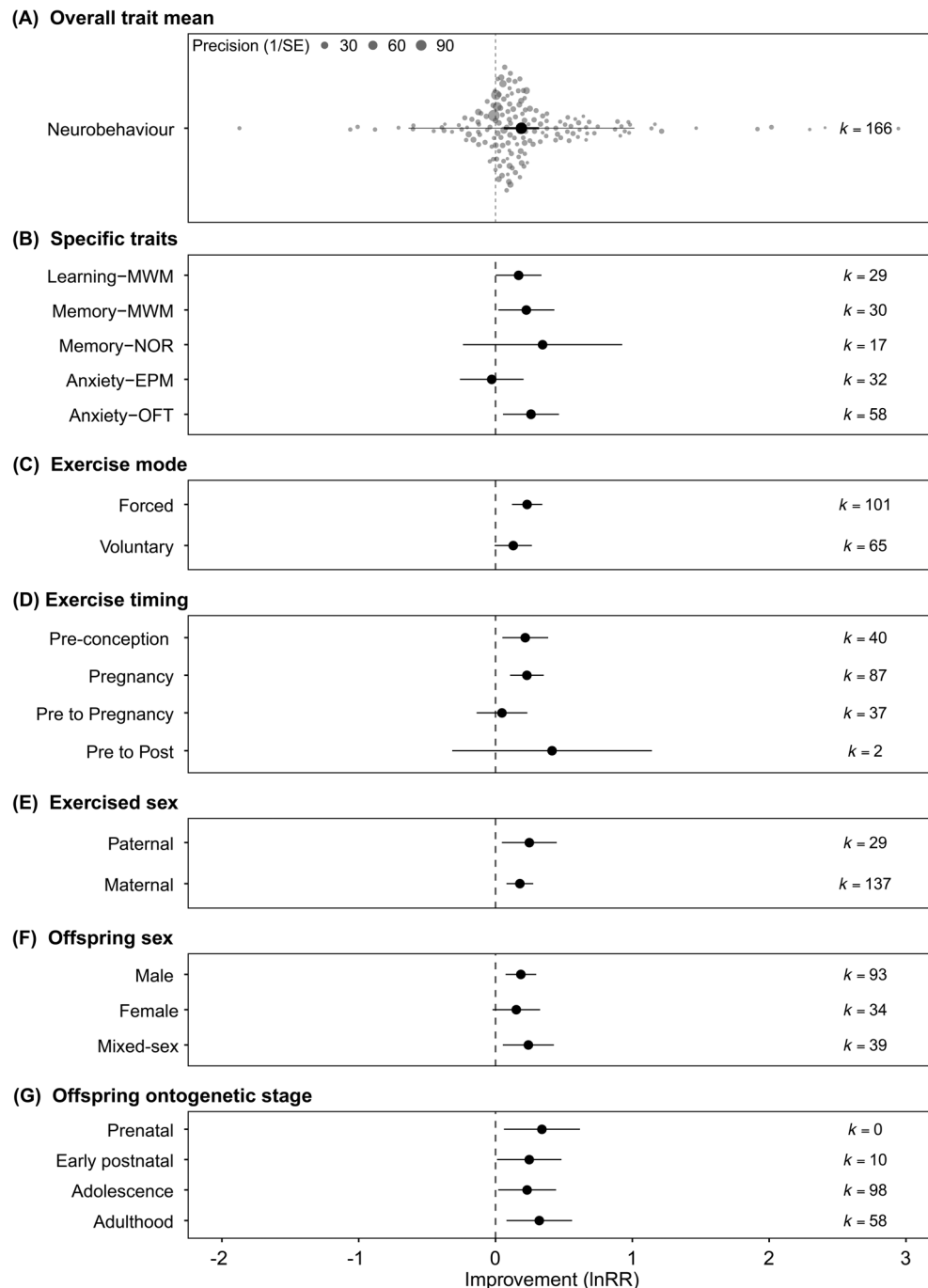


Fig 5. Meta-analytic and meta-regression results for neurobehaviour improvement and its more specific trait categories (learning, memory and anxiety, split by test type). (A) Orchard (forest-like) plot displaying the overall effect of parental exercise on offspring neurobehaviour. (B) Forest plots showing trait variability within different trait categories. MWM, Morris water maze; EPM, elevated plus maze; NOR, novel object recognition; OFT, open field test. (C–G) Forest plots displaying neurobehaviour effect size estimates (lnRR) from univariate regression models with potentially influential moderators: (C) parental exercise mode; (D) parental exercise timing (Pre to Pregnancy, pre-conception to pregnancy; Pre to Post, pre-conception to *post-partum*); (E) exercised sex (parental sex); (F) offspring sex; (G) offspring ontogenetic stage. Positive values of lnRR imply that parental exercise improves offspring brain neurobehaviour. Other details are as in legend to Fig. 3.

(b) Overall trait variation

Parental exercise did not significantly reduce inter-individual differences in brain structure (5.2%, $\beta_{\ln CVR} = -0.0536$,

CI = -0.402 to 0.294 , $k = 95$, Fig. 6A and Table S12). However, we found that parental exercise significantly reduced inter-individual differences in offspring's neurobehaviour by 10.2% ($\beta_{\ln CVR} = -0.107$, CI = -0.196 to -0.018 ,

$k = 166$, Fig. 7A and Table S13), showing that parental exercise affects all offspring in a similar way. There was a generally high degree of total heterogeneity ($I^2 = 74.8\text{--}99.9\%$ depending on the trait; Tables S18–S20).

(3) Forced or voluntary exercise of the parent?

Forced parental exercise significantly enhanced offspring brain structure and neurobehaviour (e.g. 28.6% increase in brain growth factor expression, $\beta_{\text{lnRR}} = 0.251$, CI = 0.037 to 0.466, $k = 79$, Fig. 3C, Table S21; 19.0% improvement in brain structure, $\beta_{\text{lnRR}} = 0.174$, CI = 0.057 to 0.291, $k = 56$, Fig. 4C, Table S21; 26.0% improvement in neurobehaviour, $\beta_{\text{lnRR}} = 0.231$, CI = 0.120 to 0.342, $k = 101$, Fig. 5C, Table S21). Forced parental exercise also significantly reduced (a 44.6% reduction) global DNA methylation in offspring hippocampus ($\beta_{\text{lnRR}} = -0.591$, CI = -1.047 to -0.136, $k = 5$, Table S14).

In contrast to our prediction, voluntary parental exercise was not as effective as forced exercise in eliciting increases in offspring brain structure (a non-significant 8.3% increase: $\beta_{\text{lnRR}} = 0.08$, CI = -0.035 to 0.194, $k = 39$, Fig. 4C, Table S21). On average, voluntary parental exercise had 9.7% smaller effects than forced exercise on brain structure (contrast model: $\beta_{\text{lnRR}} = -0.0944$, CI = -0.183 to -0.006, $k = 39$, Table S22). However, we found no significant difference for brain growth factor expression (contrast model: $\beta_{\text{lnRR}} = 0.016$, CI = -0.174 to 0.206, $k = 52$, Table S22). Interestingly, forced parental exercise significantly reduced inter-individual differences by 25.7% in brain structure ($\beta_{\text{lnCVR}} = -0.297$, CI = -0.520 to -0.074, $k = 56$, Fig. 6C and Table S21) and by 14.2% in neurobehaviour ($\beta_{\text{lnCVR}} = -0.153$, CI = -0.268 to -0.039, $k = 101$, Fig. 7C, Table S21). We also found that forced parental exercise had a significantly larger effect than voluntary exercise on inter-individual differences in offspring cognition-related traits (e.g. 52.9% reduced variability in brain structure; contrast: $\beta_{\text{lnCVR}} = 0.425$, CI = 0.102 to 0.747, $k = 39$, Table S22).

(4) Short-duration or long-duration exercise?

Overall, parental exercise that was of short duration (e.g. pre-conception alone) had a small, but statistically significant, effect on brain growth factor, brain structure and neurobehaviour (Table S23; brain growth factor: $\beta_{\text{lnRR}} = 0.238$, CI = 0.032 to 0.444, $k = 45$, Fig. 3D; brain structure: $\beta_{\text{lnRR}} = 0.178$, CI = 0.02 to 0.336, $k = 15$, Fig. 4D; neurobehaviour: $\beta_{\text{lnRR}} = 0.217$, CI = 0.050 to 0.385, $k = 40$, Fig. 5D). For parents exercised over a longer period (pre-conception to pregnancy, pre-conception to *post-partum*) no statistically significant effects were found on offspring traits of interest (Figs 3D–5D and Table S23). However, we did not find any significant differences between short and longer periods of exercise on brain growth factor, brain structure and neurobehaviour (see Table S24).

The parallel analysis of variation indicated a tendency for short-duration exercise during pregnancy to reduce inter-individual differences in brain structure and neurobehaviour

performance (near-significant decrease of 20.3% in brain structure, $\beta_{\text{lnCVR}} = -0.228$, CI = -0.493 to 0.038, $k = 56$, Fig. 6D and Table S23; near-significant decrease of 8.9% in neurobehaviour, $\beta_{\text{lnCVR}} = -0.094$, CI = -0.221 to 0.033, $k = 87$, Fig. 7D and Table S23).

(5) Maternal or paternal exercise?

Both paternal exercise and maternal exercise significantly increased neurobehaviour performance (paternal: $\beta_{\text{lnRR}} = 0.247$, CI = 0.047 to 0.447, $k = 29$; maternal: $\beta_{\text{lnRR}} = 0.178$, CI = 0.08 to 0.275, $k = 137$; Fig. 3E and Table S25). However, paternal exercise only had a non-significant 7.1% larger effect compared to offspring where only mothers were exercised (contrast: $\beta_{\text{lnRR}} = -0.069$, CI = -0.291 to 0.153, $k = 137$, Fig. 3E and Table S26). By contrast, brain structure and neurobehaviour did not differ significantly depending on whether mothers or fathers were exercised (contrast between mothers and fathers in brain structure: 1.2%, $\beta_{\text{lnRR}} = 0.012$, CI = -0.149 to 0.173, $k = 89$, Fig. 4E, and Table S26; contrast between mothers and fathers in neurobehaviour: -6.7%, $\beta_{\text{lnRR}} = -0.069$, CI = -0.291 to 0.153, $k = 137$, Fig. 5E, and Table S26).

Notably, we found that maternal exercise significantly reduced inter-individual differences in neurobehaviour by 10.5% ($\beta_{\text{lnCVR}} = -0.111$, CI = -0.209 to -0.013, $k = 137$, Fig. 7E and Table S25), whereas maternal exercise showed a non-significant effect on inter-individual differences in brain structure (9.1%: $\beta_{\text{lnCVR}} = -0.087$, CI = -0.284 to 0.11, $k = 89$, Fig. 7E and Table S25). Paternal exercise had non-significant effects on inter-individual differences in neurobehaviour ($\beta_{\text{lnCVR}} = -0.09$, CI = -0.302 to 0.121, $k = 29$, Fig. 7E and Table S25) and brain structure ($\beta_{\text{lnCVR}} = -0.286$, CI = -0.869 to 0.297, $k = 6$, Fig. 7E and Table S25). Further, inter-individual variability for neurobehaviour and brain structure did not vary depending on whether the exercise regime was applied to mothers or fathers (contrast between mothers and fathers in brain structure: 22.0%, $\beta_{\text{lnCVR}} = 0.199$, CI = -0.407 to 0.805, $k = 89$, Table S26; contrast between mothers and fathers in neurobehaviour: $\beta_{\text{lnCVR}} = -0.021$, CI = -0.254 to 0.212, $k = 137$, Table S26).

(6) Son or daughter?

Parental exercise significantly increased brain growth factor and neurobehaviour for male offspring (brain growth factors $\beta_{\text{lnRR}} = 0.236$, CI = 0.047 to 0.425, $k = 83$; neurobehaviour $\beta_{\text{lnRR}} = 0.185$, CI = 0.074 to 0.297, $k = 93$; Table S27) but not for female offspring (brain growth factors $\beta_{\text{lnRR}} = 0.215$, CI = -0.02 to 0.449, $k = 11$; neurobehaviour $\beta_{\text{lnRR}} = 0.152$, CI = -0.022 to 0.326, $k = 34$; Table S27). The effects of parental exercise did not impact male and female offspring significantly differently (contrast between offspring sexes: brain growth factors $\beta_{\text{lnRR}} = -0.021$, CI = -0.179 to 0.137, $k = 11$; brain structure $\beta_{\text{lnRR}} = -0.022$, CI = -0.098 to 0.053, $k = 19$; neurobehaviour $\beta_{\text{lnRR}} = -0.034$, CI = -0.224 to 0.157, $k = 34$; Table S28).

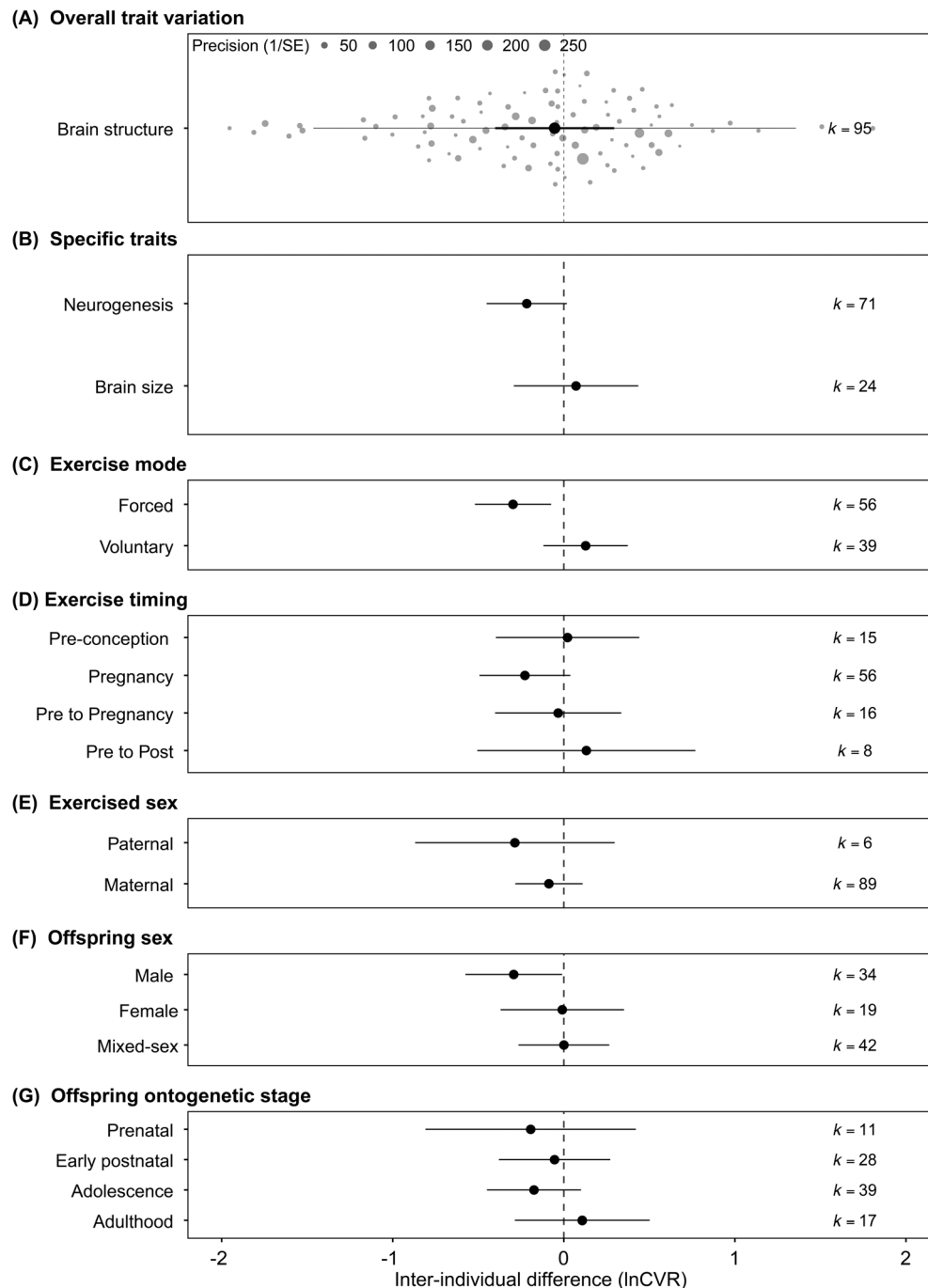


Fig 6. Meta-analytic and meta-regression results for variability in brain structure its more specific trait categories (brain size and neurogenesis). (A) Orchard (forest-like) plot displaying the overall effect of parental exercise on offspring brain structure variability. (B) Forest plots showing trait variability within different trait categories. (C–G) Forest plots displaying variability in brain structure effect size estimates (lnCVR) from univariate regression models with potentially influential moderators: (C) parental exercise mode; (D) parental exercise timing (Pre to Pregnancy, pre-conception to pregnancy; Pre to Post, pre-conception to *post-partum*); exercised sex (parental sex); (F) offspring sex; (G) offspring ontogenetic stage. lnCVR represents the differences in the coefficients of variation (i.e. SD divided by mean) in brain structure for the exercised and control groups. Negative values of lnCVR imply less variability in brain structure, i.e. parental exercise reduces inter-individual differences in brain structure. Other details are as in legend to Fig. 3.

However, parental exercise tended to reduce inter-individual differences in male cognitive traits, but not in females, with the difference between offspring sexes

being near-significant (differences: 32.8% in brain structure $\beta_{\text{lnCVR}} = 0.284$, CI = -0.074 to 0.642 , $k = 53$, Fig. 6F and Table S28; 22.2% in neurobehaviour

$\beta_{\ln\text{CVR}} = 0.201$, CI = -0.020 to 0.421 , $k = 34$, Fig. 7F and Table S28).

(7) Short-lasting or long-lasting effects of exercise?

We found a 40.4 and 26.9% increase in brain growth factor expression in both prenatal and adolescence offspring age groups, respectively (prenatal: $\beta_{\ln\text{RR}} = 0.339$, CI = 0.09 to 0.617 , $k = 14$; adolescence: $\beta_{\ln\text{RR}} = 0.238$, CI = 0.032 to 0.444 , $k = 81$; Fig. 3G and Table S29). Adulthood offspring age group showed a non-significant effect on brain growth factor expression ($\beta_{\ln\text{RR}} = 0.222$, CI = -0.284 to 0.728 , $k = 19$; Fig. 3G and Table S29). However, the adulthood offspring age group showed significantly increased brain structure (15.8%, $\beta_{\ln\text{RR}} = 0.17$, CI = 0.022 to 0.319 , $k = 17$; Fig. 4G and Table S29) and neurobehaviour (24.0%, $\beta_{\ln\text{RR}} = 0.215$, CI = 0.067 to 0.364 , $k = 58$; Fig. 5G and Table S29). Notably, our analyses did not reveal statistically significant differences in the magnitude of mean effect size among the offspring age groups (Table S30), suggesting that parental exercise exerted long-lasting benefits on offspring brain structure and neurobehaviour.

(8) Publication bias

A normal funnel plot assumes homogeneity (i.e. $I^2 = 0$), which was not the case in our data sets. Visual and statistical tests for bias in our data set, after controlling for heterogeneity (Tables S31–S42), revealed no obvious signs of publication bias, except for a slight bias in neurobehaviour performance ($\ln\text{RR}$) and brain growth factor upregulation ($\ln\text{RR}$). We present funnel plots of effect size distributions and their sampling errors and results of our Egger's regression tests in Figs S5–S16. We also did not observe much evidence for time-lag effects (Figs S17–S22), except for the brain growth factors $\ln\text{RR}$ data set. Therefore, our estimates seem to be largely free from potential publication bias.

(9) Between-study correlations

As expected, we found that effect sizes representing changes in neurobehaviour, brain structure, and gene expression of brain growth factors, were moderately correlated with each other (Pearson $r = 0.16$ – 0.62 , Figs S23–S28). In one case, the correlation was significant: increased expression of brain growth factor was significantly correlated with improvements in brain structure (Pearson $r = 0.62$, CI = 0.10 to 0.87 , $t = 2.60$, $df = 11$; Fig. S23).

IV. DISCUSSION

(1) Parental exercise provides widespread benefits for offspring brain development and cognition

(a) Improved cognitive performance

Despite mounting empirical evidence suggesting beneficial effects of parental exercise on offspring brain development

and neurobehaviour, we still know very little about the generality of intergenerational inheritance. Our main aim was to conduct the first quantitative synthesis on whether parental exercise can have positive effects on F1 offspring traits related to brain and neurobehaviour. The available empirical evidence indicates that parental exercise significantly increased offspring brain growth factor expression by 29.1% (Fig. 3A), showed a tendency to increase brain structure by 12.7% (although statistically non-significant, Fig. 4A), and significantly enhanced neurobehaviour performance by 20.8% (Fig. 5A).

(b) Reduced variation in cognition

Using meta-analyses of variation (Nakagawa *et al.*, 2015) we observed that parental exercise significantly reduced inter-individual variability in neurobehaviour by 10.2% (Fig. 7A) when controlling for trait mean differences (i.e. using $\ln\text{CVR}$). This result suggests that offspring from the exercised parent(s) had more stable neurobehavioural performance (less variation) than those from control parents. Note, however, that there is no reason to assume that parental exercise will be homogeneously effective for all offspring because when means differ, variances around the means usually change in the same direction (i.e. mean–variance relationships, also known as Taylor's power law) (O'Dea *et al.*, 2018, 2019; Nakagawa *et al.*, 2015).

(c) An integrated view of exercise effects

We observed an overall mean increase/improvement in offspring brain and neurobehavioural traits along with a reduction in variation. The reduction in variation suggests that parental exercise tended to benefit offspring with poor cognitive abilities much more than those with high cognitive abilities (due to a ceiling for the highest possible cognition performance values in the measured traits) (see Fig. S29 for explanation). From a statistical perspective, this is an individual $\times$ intervention interaction. For neurobehaviour, parental exercise significantly reduced inter-individual differences by 10.2% (Fig. 7A), which can be seen as reduced variability in the exercised group (Fig. S29). This reduced variability (or increased stability) is likely driven by the larger benefits of forced parental exercise than voluntary exercise (see Figs 5C and 7C).

(2) Sources of heterogeneity in heritable benefits of exercise

Because individual and exercise characteristics varied across studies, we used five main covariates (i.e. moderators) to account for heterogeneity in the observed effects (see Figs 3–5): (i) exercise mode (Akhavan *et al.*, 2008); (ii) exercise timing (i.e. exercise duration) (O'Callaghan, Ohle & Kelly, 2007); (iii) sex of the exercised parent (Robinson & Bucci, 2014); (iv) offspring sex (Short *et al.*, 2017); and (v) offspring ontogenetic stage (Ferrari *et al.*, 2018). Most of these

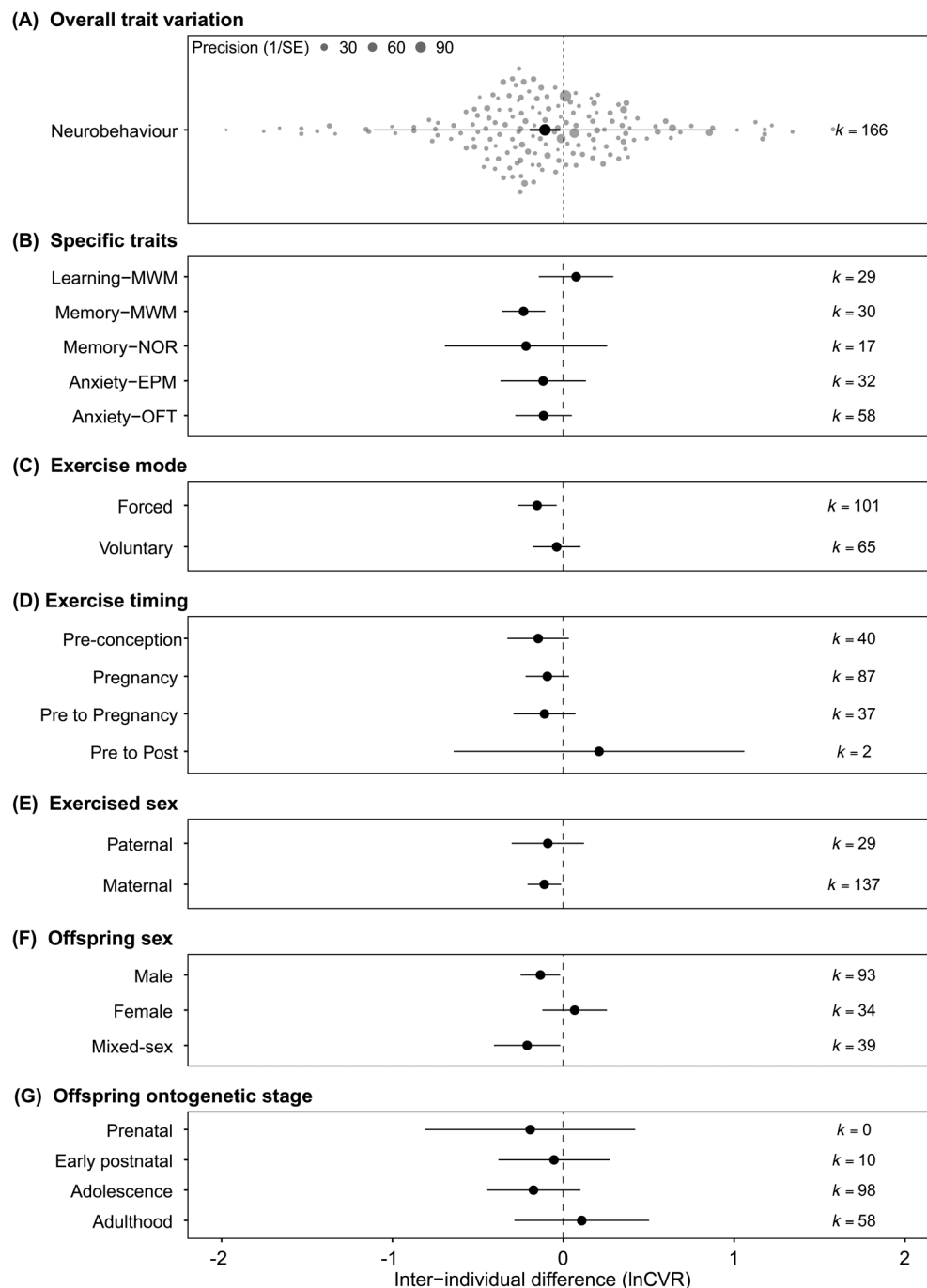


Fig 7. Meta-analytic and meta-regression results for variability in neurobehaviour and its more specific trait categories (learning, memory and anxiety, split by test type). (A) Orchard (forest-like) plot displaying the overall effect of parental exercise on offspring neurobehaviour variability. (B) Forest plots showing trait variability within different trait categories. MWM, Morris water maze; EPM, elevated plus maze; NOR, novel object recognition; OFT, open field test. (C–G) Forest plots displaying neurobehaviour variability effect size estimates (lnCVR) from univariate regression models with potentially influential moderators: (C) parental exercise mode; (D) parental exercise timing (Pre to Pregnancy, pre-conception to pregnancy; Pre to Post, pre-conception to *post-partum*); (E) exercised sex (parental sex); (F) offspring sex; (G) offspring ontogenetic stage. Negative values of lnCVR imply less variability in neurobehaviour, i.e. parental exercise reduces inter-individual differences in neurobehaviour. Other details are as in legends to Figs 3 and 6.

covariates accounted for low to moderate amounts of heterogeneity in the effects (general trait category $R^2 = 0\text{--}7.6\%$; specific trait $R^2 = 0\text{--}50.3\%$) (Tables S43–S48). We discuss

below how these covariates moderate the intergenerational benefits of parental exercise and provide a discussion on what moderators might explain the substantial heterogeneity in

these data that we were not able to account for in our analyses.

(a) Exercise type

Forced exercise (e.g. treadmill running, forced swimming) serves as an “intense stressor” (Yanagita *et al.*, 2007, p.356), which is predicted to have fewer positive effects on offspring cognition compared to voluntary exercise. Surprisingly, forced exercise significantly improved brain structure by 19.0% (Fig. 4C), and voluntary exercise had a 9.7% smaller effect than forced exercise (Table S22). More notably, forced exercise significantly reduced the variability in brain structure, by 52.9% (Fig. 6C; Table S22). We observed similar, although weaker, patterns for brain growth factors and neurobehaviour. There are four potential explanations for these intriguing results. First, voluntary exercise ignored the existence of intra-individual variation in parents (e.g. trainability, amount of training performed) subjected to voluntary exercise (i.e. free-running wheel); a point that highlights limitations in the protocols used to examine the intergenerational effects of exercise. Locomotion in voluntary exercise is spontaneous and not directly motivated by external stimuli (Garland *et al.*, 2011), with total travel distance being the exclusive measure of voluntary exercise used (Bick-Sander *et al.*, 2006; Kim *et al.*, 2007). Some published work indicates large differences in individual motivation to use a running wheel, and subjects were classified into three categories according to activity-related measurements: inactive, active, and very active (Amorim *et al.*, 2009; Santana Muniz *et al.*, 2014; Fragoso *et al.*, 2017). In the majority of other voluntary exercise studies, data for all animals were pooled together, potentially diluting the effects of exercise and increasing variability.

Second, prenatal stress may be an important factor that can mediate effects on offspring (Fujioka *et al.*, 2006). A number of studies revealed that short-lasting parental or maternal stressors could improve hippocampal development and cognition in offspring (Cannizzaro *et al.*, 2006; Fujioka *et al.*, 2006; Soares-Cunha *et al.*, 2018). Empirical studies usually subject rodents to forced exercise only for a short period each day (20–60 min/day for treadmill, 10–60 min/day for swimming in our data set). Thus, it is possible that short-term forced exercise serves as a mild prenatal stressor, improving, rather than reducing, offspring brain development and cognitive performance.

Third, physical exercise during the inactive phase of a daily cycle might enhance rodent neurogenesis, whereas exercise during the active phase could suppress it (Holmes *et al.*, 2004). This may be consistent with our results: because rodents are nocturnal, voluntary exercise is likely to occur in the dark phase, while forced exercise usually takes place in the light phase (i.e. when rodents are normally inactive).

Fourth, from a perspective of energy equilibrium, mothers require extra energy during pregnancy and lactation. Therefore, pregnant mothers may prioritize the available energy for reproduction over other body functions, such as physical

activity and exercise. Voluntary exercise might therefore be kept to a minimum during these periods, making any changes to offspring brain function and structure difficult to observe. In addition, mothers with fewer energy resources might spend less time in voluntary exercise than those with sufficient energy reserves.

In addition, we found an interesting effect of the moderators of exercise type. Within a voluntary exercise protocol, an enriched environment is suggested to have a greater effect than wheel running in an unenriched environment: an enriched environment resulted in a more complex intergenerational effect (Bohacek & Mansuy, 2015). The benefits on offspring cognitive function were not solely due to physical activity, with other associated effects suggested to contribute to these benefits, including the differential allocation hypothesis (Burley, 1988) and paternal social enrichment (Leuner, Glasper & Gould, 2010; Mashoodh *et al.*, 2012). However, we found: (i) non-significant effects on offspring brain structure and neurobehaviour both in an enriched environment (brain structure: $\beta_{\text{lnRR}} = 0.109$, CI = -0.08 to 0.298 , $k = 12$; neurobehaviour: $\beta_{\text{lnRR}} = 0.143$, CI = -0.016 to 0.303 , $k = 44$; Table S49) and wheel running (brain structure: $\beta_{\text{lnRR}} = 0.054$, CI = -0.077 to 0.186 , $k = 27$; neurobehaviour: $\beta_{\text{lnRR}} = 0.091$, CI = -0.114 to 0.296 , $k = 21$; Table S49); (ii) an enriched environment did not show statistically significant differences from wheel-running for offspring traits related to brain and neurobehaviour (contrast between enriched environment and wheel-running in offspring brain growth factors: $\beta_{\text{lnRR}} = 0.059$, CI = -0.297 to 0.415 , $k = 30$; contrast in brain structure: $\beta_{\text{lnRR}} = -0.006$, CI = -0.215 to 0.204 , $k = 27$; contrast in neurobehaviour: $\beta_{\text{lnRR}} = -0.066$, CI = -0.299 to 0.168 , $k = 21$; Table S50).

(b) Sex differences

In our data set, we had two forms of sex-specific effects: (i) the sex of the exercised parent (maternal or paternal exercise), and (ii) the sex of the examined offspring (male or female progeny). We predicted that maternal exercise would result in greater improvements to offspring cognition than paternal exercise because the ovum has a substantially larger size than sperm and can contribute more non-genetic/epigenetic cytoplasmic factors to offspring (Cosmides & Tooby, 1981). Further, in mammals, mothers provide substantial maternal care during pregnancy and lactation (Vik *et al.*, 2014), whereas fathers merely contribute the sperm that is needed for fertilization (Crean & Bonduriansky, 2014).

Maternal exercise had a 22.2% larger positive effect, albeit non-significant, on the expression of brain growth factors than paternal exercise (Fig. 3E). However, maternal and paternal exercise seemed to contribute similarly to offspring brain development and cognitive improvement (Table S25). This finding suggests that fathers were able to transmit their experience of exercise, likely through the ejaculate. This is in line with recent results showing paternal transmission of (intrinsic and extrinsic) environmental effects

via small non-coding RNAs in sperm (Short *et al.*, 2017; Benito *et al.*, 2018; McGreevy *et al.*, 2019).

With regard to offspring sex, previous studies reported that males are more susceptible to gestational stress/stimuli during development (Mueller & Bale, 2007; Robinson & Bucci, 2014). Therefore, we predicted that male offspring would exhibit greater benefits of parental exercise than female offspring. We found no statistically significant differences between the effects on male and female offspring, in terms of both mean trait values and trait variabilities (Figs 3F–7F).

(c) Temporal effects

We examined two types of timing-related moderators: (i) developmental exposure windows of parental exercise, and (ii) ontogenetic stages of offspring at measurement. Our analysis implies that longer-term exercise (pre-conception to pregnancy, pre-conception to *post-partum*) did not translate into better cognition-related performance in their offspring. Ludyga *et al.* (2020) showed that long-duration exercise seemed to induce only slight improvements in the cognition of healthy animals in comparison to individuals with cognitive deficits. It is also possible that long-duration exercise might even serve as a form of prenatal stress to foetal development. Prenatal stress can lead to negative physiological (Mastorci *et al.*, 2009), neurodevelopmental (Graginc-Philippe *et al.*, 2014), and behavioural alterations (Kinney *et al.*, 2008) in the offspring. The benefits of parental exercise benefits are expected to gradually diminish as offspring grow from postnatal stages to adulthood (Salthouse & Davis, 2006). By contrast, we found that adult offspring still showed significant benefits of parental exercise in brain structure, and neurobehaviour (15.8, and 24.0%, respectively; Table S29). These findings, importantly, suggest that changes to brain structure and function induced at early stages of development remain beneficial throughout life.

(d) Unexplained heterogeneity

Despite testing the effects of moderators regarding experimental design and biological characteristics (see Section IV.2), there was still much heterogeneity that remained unexplained ($1 - R^2 = 49.7$ to 100% depending on the trait) (Tables S43–S48). Our data set generally exhibited a high degree of total heterogeneity (i.e. effect heterogeneity excluding sampling error variance) both for effects on the mean (lnRR: $I^2 = 95.2$ –98.3%, Tables S15–S17) and variance (lnCVR: $I^2 = 74.8$ –99.9%; Tables S18–S20). When decomposing the total variance in our multilevel models (see I^2 values for each level; Tables S15–S20), the heterogeneity due to study identity (I^2 between-study) is usually relatively high comparing to the other levels of random effect (i.e. trait category, strain, or residual heterogeneity). For example, the between-study heterogeneity (I^2) is 49.9% for brain growth factors (lnRR, Table S15), which can explain 52.4% of the total heterogeneity (total $I^2 = 95.2\%$). For the specific types of brain growth factors, the between-study

heterogeneity (I^2) is even higher (e.g. $I^2 = 72.4\%$ in reelin, Table S15). This high degree of unexplained heterogeneity in our analyses suggests that we might have missed some potentially important moderators. Quantitative factors, such as exercise intensity or frequency, could be such moderators: human studies show a dose response between exercise intensity and cognitive function (Ludyga *et al.*, 2020). We did attempt to extract these qualitative parameters (see data sets at https://github.com/Yefeng0920/Intergenerational_effect_of_exercise_MA_Git.) However, many studies reported the relevant information in incompatible units (e.g. speed, duration, distance, time), which prevented us from modelling these quantitative factors.

(3) Meta-analytic insights into neural mechanisms

Our meta-analyses on growth factor expression in offspring brain provide insight into potential mechanisms underlying the benefits of parental exercise on offspring brain development and cognition (Fig. 8). Especially noteworthy are four factors expressed in the brain: (i) hippocampal neurogenesis; (ii) VEGF signalling; (iii) BDNF signalling; and (iv) epigenetic modifications.

(a) Hippocampal neurogenesis

Parental exercise significantly increased hippocampal neurogenesis in offspring by 16.5% (Fig. 4B and Table S12). Neurogenesis occurs throughout ontogeny, including during the early postnatal stage (Dayi *et al.*, 2012) and adolescence (Segabinazi *et al.*, 2019). Thus, even adult offspring from exercised parents could have greater numbers of new neurons in the hippocampus (McGreevy *et al.*, 2019).

(b) VEGF signalling

Offspring from exercised parents exhibited significantly greater VEGF levels (i.e. 35.8% increase) compared to offspring from sedentary parents (Table S11). VEGF is a growth factor that is linked to vascular endothelial cell formation and development in the brain (Lopez-Lopez, LeRoith & Torres-Aleman, 2004). Increased VEGF expression could facilitate angiogenesis in the offspring brain, providing the nutrients and oxygen necessary to support brain development and function, including neurogenesis in the hippocampus (Bick-Sander *et al.*, 2006) and cortex (Sanches *et al.*, 2018).

(c) BDNF signalling

Parental exercise also significantly upregulated the release of BDNF in offspring brain by 28.9%, on average (Table S11). BDNF plays a central role in brain function and cognition (Kuipers & Bramham, 2006). The use of antibodies blocking the functions of the tyrosine kinase receptor (TrkB, the receptor for BDNF) indicate that BDNF signalling is necessary for exercise-induced neurogenesis and improved hippocampal-dependent learning and memory (Vaynman *et al.*, 2006). Our results also show that parental exercise resulted in

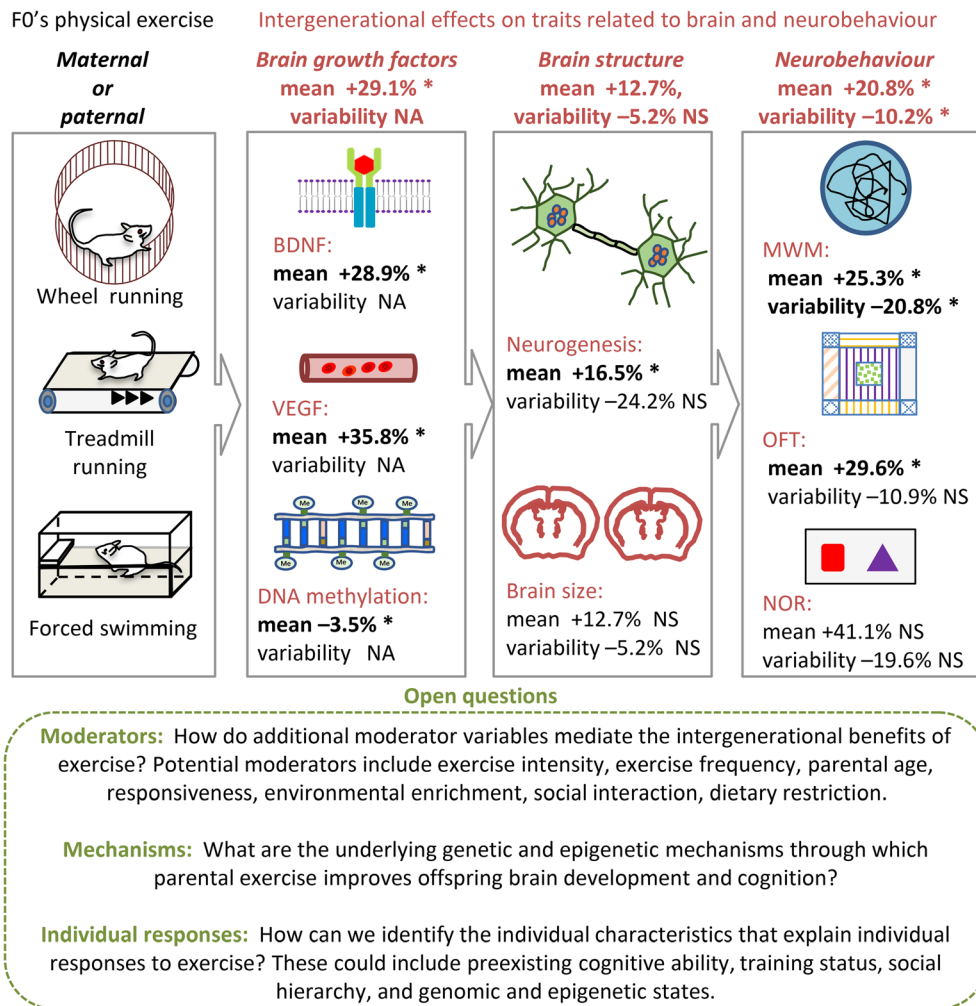


Fig 8. Meta-analytic insights into intergenerational effects of physical exercise on traits related to brain and neurobehaviour. VEGF signalling, BDNF signalling, and epigenetic modification can mediate the transmission of positive effects of physical exercise on traits related to brain and cognition. The positive correlation between expression of brain growth factors and brain structure implies that parental exercise improves brain structure *via* upregulating brain growth factors and epigenetic modifications. Changes in brain structure might translate into measurable improvements in neurobehaviour (improved learning and memory, and reduced anxiety). Apart from improved cognitive performance (the effect on mean trait values), parental exercise reduced inter-individual differences in offspring neurobehaviour (the effect on variability). This means that parental exercise tended to benefit individuals with poor cognitive abilities much more than those with high cognitive abilities (ceiling of highest possible cognitive performance). However, some potential moderator variables could modulate the positive effects of exercise. The identified moderators included biological characteristics (parental sex, offspring sex and age), qualitative (exercise mode), and quantitative exercise characteristics (exercise duration). This schematic summary is far from complete. Many questions remain unresolved, such as other potential moderators (e.g. exercise intensity, social interaction). *, P -value < 0.05; NS, non-significant (P -value > 0.05); NA denotes that an accurate estimate of variability in gene expression is unavailable due to a lack of information about reference genes. We presented the detailed reasons in the main text and Table S11.

significantly increased secretion of TrkB in the hippocampus by 14.2% (Table S11). Parental exercise also tended to enhance the expression of synaptic protein (i.e. synapsin) in offspring brains, with these factors increasing by 11.5% (Table S11). Importantly, our results confirm that the expression of brain growth factors (e.g. BDNF and VEGF) is positively related to enhanced brain structure (i.e. neurogenesis) (Fig. S23), indicating that parental exercise-dependent activation of BDNF signalling is linked to improved offspring

brain development and cognition. In addition to improvements in traits related to cognition, our results also show that parental exercise significantly reduced offspring anxiety (Fig. 5B). Reduced offspring anxiety may be linked to the increased expression of BDNF elicited by parental exercise, i.e. BDNF could be involved in the observed protective effects of parental exercise on offspring anxiety. In humans, BDNF deficit is linked with mood disorders such as depression (Cotman *et al.*, 2007).

(d) Epigenetic modifications

The effects of parental exercise on the transcription of brain growth factors in offspring are potentially mediated by epigenetic mechanisms, such as DNA methylation, histone acetylation, and non-coding RNAs (Sleiman *et al.*, 2016). This opens the possibility that epigenetic modifications resulting from parental exercise are heritable and regulate gene expression of growth factors in offspring (e.g. BDNF and VEGF). Parental exercise significantly decreased global DNA methylation in the offspring hippocampus by 3.5% ($\beta_{\ln RR} = -0.035$, CI = -0.049 to -0.022 ; Table S14). Note that this decreased global DNA methylation should be interpreted with caution because the model was fitted only using four studies on 78 animals. In addition, a few studies in our data set report that non-coding RNAs (microRNA) act as potent epigenetic mediators for brain function and cognition (Short *et al.*, 2017; Benito *et al.*, 2018; McGreevy *et al.*, 2019).

(4) Limitations and future directions*(a) Limitations*

Although we examined cohort (sex and age), qualitative (exercise modes) and quantitative characteristics (exercise duration) as potential moderators to account for heterogeneity, there was still much unexplained heterogeneity (see Section IV.2*d*). This warrants the exploration of further moderators (Fig. 8). For example, a high exercise intensity tends to predict better cognition performance than lower intensity exercise in humans (Ludyga *et al.*, 2020). In our analyses, we could not quantify exercise intensity (see Section IV.2*d*). Future studies should attempt to address additional moderator variables potentially mediating the intergenerational benefits of exercise, and better reporting of full experimental details will aid future synthesis efforts.

(b) Future directions

Based on our meta-analytic results, we have five recommendations for future empirical studies. First, we recommend employing forced exercise protocols (e.g. treadmill, swimming) rather than voluntary exercise (e.g. running wheel), to minimize intra-individual variability. For studies investigating the mechanisms underlying transgenerational inheritance, forced exercise ensures similar effort among individuals. Because forced exercise can minimize intra-individual variability, researchers can reasonably expect that exercised individuals will exhibit positive effects of exercise. However, voluntary exercise may be more appropriate to probe what type of regimes might be most suitable for driving exercise-induced effects. The observed high heterogeneity ($\ln RR$: $I^2 = 95.2-98.3\%$, $\ln CVR$: $I^2 = 74.8-99.9\%$) suggests that the approaches used show a wide variation in their effectiveness across studies, and that future work should attempt to identify the most effective exercise regimes and to understand why they work best. It may also be worth investigating whether a combination of forced and voluntary exercise is

most effective in improving brain development and cognition. Future studies should quantify exercise parameters for each individual according to cardiovascular fitness measures, such as individual maximal oxygen intake (VO_{2peak}) (Holzschnieder *et al.*, 2012).

Second, empirical evidence indicates that dietary restriction is beneficial for cognition (Witte *et al.*, 2009), whereas social isolation can undermine the benefits of exercise on brain development (Stranahan, Khalil & Gould, 2006). Such research suggests potential interactions between exercise and other lifestyle and environmental factors, as well as interactions between maternal and paternal exercise. Unfortunately, our literature search located only two studies that measured the effects of these interactions (Molteni *et al.*, 2004; Bechara & Kelly, 2013). This would be an interesting direction to pursue in future empirical and meta-analytical studies.

Third, our literature search did not find studies that measured transgenerational effects of exercise on cognition for F2 and later generations. Recent evidence indicates that transgenerational inheritance could skip the F1 generation and instead transmit benefits to later descendants (Yeshurun *et al.*, 2017). Therefore, we appeal for empirical studies examining the transgenerational transmission of exercise effects to the F2 generation and beyond.

Fourth, we discussed meta-analytic insights into the neural mechanisms underlying parental exercise effects (see Section IV.3). While our mechanistic claims are intriguing, there is still much to be learned. Clearly, we need intricate empirical experiments to validate these insights into the neural mechanism underlying the effects of exercise. Future research should attempt to elucidate changes in the regulatory network of signal transduction, gene transcription and protein expression to validate our meta-analytic insights. For example, such work could attempt to unravel the mechanisms by which non-coding RNAs (microRNA) transmit benefits of exercise into the next generation.

Finally, our meta-analysis highlights the importance of inter-individual variability in responses to exercise interventions. The differences we observed in individual responses to treatments are likely to have biological importance. For example, they imply the existence of among-individual heterogeneity of treatment effects – a core idea in stratified, precision or personalised medicine (Senn, 2018). Future studies should employ powerful statistical methodology (e.g. meta-analysis of variation) to interpret individual responses to interventions, to meet the prerequisites of such approaches. High-throughput next-generation sequencing technology provides a convenient and rapid approach to quantifying individual (epi-)genetic characteristics with which researchers could compare individual (phenotypic) responses.

V. CONCLUSIONS

- (1) We show that parental physical exercise helps to enhance offspring brain growth factor expression

(i.e. BDNF, TrkB, VEGF), brain structure (neurogenesis), and neurobehaviour (learning, memory, and anxiety). Moreover, parental exercise significantly reduces variability in offspring neurobehaviour (on average 10.2%), suggesting there is an individual $\times$ exercise interaction. Meta-analysis of variance alongside means allowed us to obtain a more integrated picture of exercise effects: parental exercise not only improves offspring neurobehavioural performance but also reduces inter-individual differences in some aspects of neurobehaviour.

- (2) Moderator analyses (univariate meta-regression) on biological characteristics, revealed sex differences in exercise effects. Both maternal and paternal exercise significantly improved their offspring's neurobehaviour, whereas only maternal exercise significantly reduced neurobehaviour variability. We found no statistically significant difference between the effects on male and female offspring, in terms of both mean trait values and trait variabilities.
- (3) Moderator analyses on intervention characteristics showed that forced exercise was more efficient than voluntary exercise at improving neurobehaviour performance (26.0%) and reducing variability (14.2%). We found that short-duration exercise (during pre-conception or pregnancy) is the best time to exercise parents to observe neurobehavioural benefits in offspring.
- (4) Our meta-analytic approach suggests a number of likely mechanisms mediating the benefits of parental exercise on offspring brain structure and cognition, including hippocampal neurogenesis, improved VEGF and BDNF signalling, and epigenetic modifications. Parental exercise significantly increases offspring's hippocampal neurogenesis from an early postnatal stage to adulthood (18.0–22.7%). It also significantly upregulates the release of VEGF (35.8%), BDNF (28.9%) and TrkB (14.2%) and decreases global DNA methylation in the hippocampus (3.5%).

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VIII. Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article.
Appendix S1. Search strings, and supplementary tables and figures.

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